Carlos Ojeda-de-Silva

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Education

Harvey Mudd College August 2023 – May 2027
Bachelor of Science in Engineering (GPA: 3.5 / 4.0) *Claremont, CA*

Experience

Lead, Electrical and Computer Engineer August 2023 – Present
HMC Liquid Rocket Engine *Claremont, CA*

- Manufacturing a liquid rocket engine that is expected to produce 1000+ lbf of thrust
- Designing and building a high-power rocket to achieve a NAR L2 certification
- Currently leading the Electrical and Software team for the Liquid Rocket Engine

Biomedical Engineer January 2025 – Present
CuTE Lab at Harvey Mudd College *Claremont, CA*

- Created a 4-channel pressure pump for precise microfluidic control
- Retrofitted a 3D printer for pressure extrusion of bioinks like sodium alginate and polyethylene glycol
- Developed a custom toolpath slicer to print arbitrary geometries while optimizing extrusion paths to minimize extruded material stiffness gradient
- Engineered a PDMS microfluidic device with specialized geometric features to control individual channel resistance and optimize fluid mixing

Electrical and Computer Engineer August 2026 – Present
Blackrock Neurotech *Claremont, CA*

- Designing a custom ESP32 board to compress and relay BCI data through Wi-Fi while minimizing power usage

Mechanical Engineer August 2025 – December 2025
Northstar Radioisotopes *Claremont, CA*

- Designed specialized maintenance tools for robotic systems to minimize worker radiation exposure
- Designed and tested a dual-function tool to disconnect pipe connectors and cut steel pipes with minimal burrs

Mechanical Engineer November 2023 – December 2024
Sokil Imaging *Claremont, CA*

- Engineered and validated lightweight mounts for magnetometers reducing mechanical oscillations and signal noise
- Patented drone sensor mounting currently deployed in Ukraine for de-mining through magnetometry

Recovery Engineer August 2021 – June 2023
Alamo Heights High School Rocketry Program *San Antonio, TX*

- Simulated and calculated the physical variables during the rocket flight
- Presented the rocket flight simulation results to NASA engineers
- Engineered for over 250 hours to manufacture and calculate the flight profile of the rocket
- Designed, built, and tested a system in order to get a 140 lb rocket to safely land from 50,000 ft
- Launched this rocket in White Sands Missile Range, reaching 18,994 ft with only a half-filled tank

Projects

Compressed Video Streaming | *Jupyter Notebook, ESP32, LoRa* May 2024 – Present

- Developed neural network autoencoder compression to reduce the packet sizes over LoRa
- Evaluate other data compression algorithms (PCA, Frequency Domain Compression) to maximize bandwidth
- Architect a pipeline to send Accelerometer, Barometer, and GPS data through the LoRa channel

Portable DJ Set | *Questa, Lattice Radiant, SEGGER* October 2025 – December 2025

- Implemented real-time audio processing at dynamic sampling by interfacing an MCU with an FPGA for DSP
- Designed a joystick controller to change playback speed and applied FIR filtering

Technical Skills

Languages: Java, Python, C/C++, System Verilog, MATLAB
Developer Tools: SolidWorks, Altium Designer, VS Code, Git, Lattice Radiant, Questa, Google Colab, Jupyter Notebook, Arduino IDE, SEGGER
Additional Skills: CAD, PCB Design, Machining, Single Board Computer, Microcontroller, FPGA, 3D printing, Laser Cutting, Finite Element Analysis